

PLANT-BASED DIET QUALITY AND PREVALENCE OF TYPE-2 DIABETES AMONG AUSTRALIAN ADULTS: A NATIONWIDE CROSS-SECTIONAL STUDY

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INTRODUCTION

Type-2 diabetes (T2D) is one of the most common metabolic illnesses worldwide [1]. Dietary factors, such as low intakes of fruits and vegetables and high intakes of processed meats, are leading risk factors for T2D [2]. Prospective and cross-sectional studies have shown that following a healthy plant-based diet (PBD) is associated with an 8-45% lower risk [3, 4] and a 12-30% lower prevalence [5, 6] of T2D, respectively. However, such evidence is lacking among the Australian population. This study aimed to investigate the cross-sectional association between PBD quality and T2D prevalence using recently released data from the 2023 National Nutrition and Physical Activity Survey.

METHODS

Dietary intake was gathered on two separate occasions using Intake24, a computerised 24-hour dietary recall system, and the Australian Food and Nutrient Database 2023 was employed to estimate energy and nutrient intakes. PBD quality was examined using the overall plant-based diet index (PDI), healthy PDI (hPDI), and unhealthy PDI (uPDI). T2D cases were defined as self-reported physician-diagnosed diabetes. Logistic regression models were used to estimate odds ratios (ORs) and 95% confidence intervals (95% CIs) for the association between a 10-point increase in PDI, hPDI, and uPDI, and T2D. Models were adjusted for socio-demographic variables (i.e., sex, age, education, country of birth, rurality and area-level disadvantage), behavioural (i.e., smoking), health-related (BMI and family history of T2D) variables and energy misreporting. Interaction with sex was tested.

RESULTS AND DISCUSSION

A total of 7,043 adults (mean $\pm$ SE: age 48.4 ± 0.3 , $50 \pm 0.01\%$ female) were included in the analysis, with 271 cases of T2D identified. The scores (mean $\pm$ SE) of PDI,

hPDI and uPDI in the overall population were 53.3 ± 0.1 , 54.1 ± 0.1 , and 55.0 ± 0.1 , respectively. No significant associations were observed between indices and T2D (PDI: OR = 1.06, $p = 0.65$, 95% CI: 0.82 – 1.36; hPDI: OR = 0.84, $p = 0.70$, 95% CI: 0.69 – 1.02; uPDI: OR = 0.91, $p = 0.52$, 95% CI: 0.70 – 1.20). In the interaction analysis, the overall product term between hPDI and sex did not reach statistical significance (OR = 1.40, $p = 0.09$, 95% CI: 0.95–2.06). However, evaluation of sex-specific strata revealed that a higher hPDI was significantly associated with a reduced prevalence of T2D among men (OR = 0.72, $p = 0.024$, 95% CI: 0.54 - 0.96), whereas no association was observed in women.

CONCLUSIONS

In this large, nationally representative sample of Australian adults, higher PBD quality was associated with a reduced prevalence of T2D among male participants, though this trend did not differ significantly from females. Future research should examine the prospective association between PBD quality and T2D risk in the Australian population and whether these dietary patterns influence T2D risk differently across sexes.

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